CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

May 2018

EC210 Analog Communication

Date: 09.05.2018, Wednesday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) State the condition for existence of the Fourier Transform. Explain Physical appreciation of the Fourier Transform with suitable example. **[07]**

Q - 2 (a) How to detect Amplitude Modulated (AM) signals using envelope detector? Explain it in brief. **[06]**

(b) Explain the function of each block in practical broadcast amplitude modulation (AM) transmitter with the help of neat diagram. **[08]**

OR

Q - 2 (a) Explain the function of each block in amplitude modulation (AM) transmitter with the help of neat diagram. **[07]**

(b) Derive the relation between the output power of an amplitude modulation (AM) transmitter and the depth of modulation. **[07]**

Q - 3 (a) Explain the working of 'filter method' of generating of single sideband (SSB) signal. **[07]**

(b) i) List out the merits of single sideband (SSB) modulation over double sideband (DSB) modulation system of transmission. **[02]**

ii) Compare frequency modulation (FM) with amplitude modulation (AM). **[05]**

OR

Q - 3 (a) Describe the block diagram of "phase shift method" for single sideband (SSB) modulation. **[07]**

(b) i) Draw the frequency spectrum of single sideband (SSB) and double sideband (DSB) signal. **[02]**

ii) What is frequency modulation? Define modulation index and deviation ratio for frequency modulation (FM). **[05]**

SECTION – II

Q - 4 (a) Differentiate directly modulated FM transmitter and indirectly modulated FM transmitter. **[07]**

Q - 5 (a) i) Why automatic gain control is desire in receiver? Explain simple AGC, delayed AGC with necessary diagram in detail. **[06]**

ii) How to demodulate frequency modulated signal using quadrature detection concept? **[04]**

(b) How do you adjust the budget of a superhetrodyne receiver? Explain with the help of an example. **[06]**

OR

Q - 5 (a) i) What is image rejection? How to overcome from it? **[06]**

ii) Describe Foster-seely detector circuit for frequency demodulation. **[04]**

(b) Draw block diagram of double conversion super heterodyne receiver and describe its each block working in detail. **[06]**

Q - 6 (a) Define the signal to noise ratio, noise temperature and noise figure. **[06]**

(b) Define Noise Factor. Derive Friss's formula. **[06]**

OR

Q - 6 (a) Explain the following terms. **[06]**

i) Flicker Noise

ii) Transit time Noise

(b) Explain any two terms. **[06]**

i) industrial noise

ii) Atmospheric noise

iii) Extraterrestrial noise

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